

Sreevatsa RAVIKANTH, MSc in BioInformatics



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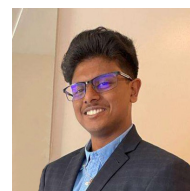
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Portfolio



Employment History

Technical experience

- Apr 2026 – Sep 2026  **Full Stack Engineer**, Districom, Montigny-lès-Cormeilles, France.
- Development of an mobile application for retail stock management for Sony France with Dart/Flutter
- Apr 2025 – Feb 2026  **Webmaster**, nutritionDay worldwide, Villejuif, France.
- Creation and management of the French branch website for the international nutritionDay study
- Adaptation of an international clinical research project to the French healthcare context
- Technical maintenance, user support, and coordination of national participation in a global epidemiological study
- Mar 2025 – Oct 2025  **Biostatistician**, Gustave Roussy, Villejuif, France.
- Contribution to scientific writing (second author of a manuscript under review)
- Statistical study on micronutrient deficiencies in head and neck cancer patients
- Construction and structuring of clinical databases for predictive modeling
- Univariate + multivariate and LASSO logistic regressions
- Apr 2024 – Sep 2024  **Biostatistician**, Gustave Roussy, Chevilly-Larue, France.
- Contribution to a scientific manuscript under review
- Data preprocessing and statistical analysis of 204 oncology patients
- Clinical data structuring for research in oncology nutrition
- Calculation of prevalences and subgroup comparisons using Boschloo's exact tests
- Identification of 6 significant micronutrients related to inflammation through Spearman correlations and symbolic regression modeling (95% CI)
- Jan 2023 – Feb 2023  **Process Optimization Consultant**, D-MUTE NETWORK, Saint-Maurice, France.
- Design and development of APIs for enterprise resource management
- Integration of Microsoft 365 licensing data into SharePoint systems

Other experience

- May 2018 – Present  **Elderly Care Assistant**, URSSAF Île-de-France, Joinville-le-Pont, France.
- Apr 2026 – Sep 2026  **Product Promoter**, Sony France / Districom, Montigny-lès-Cormeilles, France.
- Dec 2021 – Aug 2022  **Sales Advisor**, Xiaomi France, Paris, France.
- Feb 2016 – Apr 2016  **Animator**, Youth and Archive Department, City of Joinville-le-Pont, France.

Education

- 2020 – 2025  **Engineering Degree (MSc equivalent)**, École Centrale d'Électronique Paris, in Computer Science & Electronics, specialization Health & Technology.
Master's Thesis: *Rapport de stage de fin d'études*.
International exchanges: *University of Verona (2024)*, *OMNES London School (2022)*.
- 2017 – 2020  **Scientific Baccalaureate**, Lycée Teilhard de Chardin, Specialization Mathematics & Earth and Life Sciences (SVT), with high honours.

Research Publications

Journal Articles

- 1 M. Guinhut et al., "Prevalence of micronutrient deficiencies in 465 patients with head and neck cancers (hnc) admitted to a rehabilitation unit," *Clinical Nutrition ESPEN*, 2025, Manuscript under review.
- 2 A. Konesalingam, S. Ravikanth, G. M. Machado, F. Belbachir, and A. Jaouadi, "Stress detection with physiological signal from wrist-worn devices," 2025, Unpublished manuscript.

Research Projects

2023 – 2025

- **Stress Detection Using Supervised Learning**, LyRIDS, Paris, France
Co-lead author of a unpublished scientific manuscript
 - Harmonization of multimodal physiological datasets (PPG, TEMP, EDA, ACC)
 - Advanced signal preprocessing: filtering, temporal and frequency feature engineering, dimensionality reduction
 - Analysis of multimodality impact and windowing strategies
 - Training and evaluation of 12 machine learning models using LOSO-CV
 - Bayesian hyperparameter optimization
 - Best performance: F1-score of 81.9% (ternary classification)
 - Model interpretability using SHAP values
 - Development of reproducible ML pipelines on Linux systems

2024

- **Single-Cell RNA-seq Analysis of Cancer Samples Using Unsupervised Learning**, University of Verona, Italy
5-month MSc research project
 - Built a reproducible scRNA-seq analysis pipeline for 10x Genomics datasets using Seurat in R and R Markdown
 - Performed quality control, normalization, Highly Variable Gene Selection, PCA, UMAP and graph-based clustering
 - Improved data quality through automated and supervised doublet detection using DoubletFinder and lineage-specific markers
 - Annotated cell populations using SingleR, CHETAH and canonical marker genes
 - Identified cluster-specific marker genes and characterized immune and stromal populations within the NSCLC tumor microenvironment
 - Investigated cell-cell communication with CellChat and lineage trajectories using Sling-shot
- **Autonomous Drone Navigation Using Reinforcement Learning**, LyRIDS, ECE Paris, France
6-month MSc research project
 - Trained a PPO (Proximal Policy Optimization) agent under the Webots simulator using Stable-Baselines3 (PyTorch backend)
 - Designed and tuned the reward function to shape navigation behaviour
 - Trained and evaluated navigation policies for autonomous drone displacement

Research Projects (continued)

- 2022  **Face Detection Using Transfer Learning**, OMNES London School, United Kingdom
3-month research project
- Built a custom dataset of 180 subjects, with face segmentation to isolate facial regions from raw images
 - Applied random rotation for data augmentation to artificially increase the dataset size
 - Retrained a pretrained ResNet101 convolutional neural network under MATLAB
 - Achieved a 90% correct detection rate
- 2020 – 2022  **ECE CUP**, LyRIDS, Paris, France
Semi-finals, ranking among the top teams out of more than 300 competing teams
- Developed an autonomous robotics platform using a ESP32
 - Implemented real-time image processing with OpenCV to localize the robot, barrels, and opponent on the playing field
 - Designed path-planning and obstacle-avoidance algorithms for autonomous navigation
 - Integrated wireless communication through MQTT for monitoring and control
 - Competed in a robotics challenge involving barrel collection and delivery while avoiding collisions with opponents and obstacles

Skills

Languages	 French (native), Tamil (native), English (C2, TOEIC 980/990), Japanese (A1), Spanish (A1)
Programming Languages	 Python, Java, C/C++, SQL, R, MATLAB/Octave, Dart/Flutter, SAS, VHDL
Data Science & AI	 Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Bayesian Optimization, Signal Processing, Image Processing, Graph Theory
Bioinformatics	 Single-cell RNA-seq, Transcriptomics, Gene Expression Profiling, Multi-omics Data Analysis, Tumor Microenvironment Analysis, Cell Annotation, Cell-Cell Communication, Trajectory Inference
Web Development	 HTML, CSS, JavaScript, PHP, MySQL, Wordpress, Hostinger
Tools & Software	 Github, Docker, Microsoft Office (Word, Excel, PowerPoint), Microsoft Power Platform (PowerApps, Power Automate, Power BI), SharePoint, L ^A T _E X, Canva, Adobe Photoshop, SolidWorks, ThinkerCAD
Clinical & Research	 Good Clinical Practice (ICH E6 R2), Compliance Regulations, Intellectual Property
Project Management	 Agile / Decisioning, Consulting, Marketing, Communication
Embedded Systems	 Embedded Linux Systems, Arduino, VHDL, Proteus, Quartus, Cisco Packet Tracer, Wireshark

Miscellaneous Experience

Awards and Achievements

- 2024  **1st place out of 567 – Best Scientific Poster.**
 **2nd place out of 38 – Inetum Hackathon.**
- 2022  **4th place out of 300 – ECE Cup Robotics Competition.**

Certifications

- 2026  **ICH E6 (R3) Good Clinical Practice**, The Global Health Network.

Miscellaneous Experience (continued)

- 2023  **ECE Python Exam**, Liora.
-  **Project Management MOOC**, Centrale Lille.